Statistical Instance-Based Pruning in Ensembles of Independent Classifiers

Daniel Hernández-Lobato, Gonzalo Martínez-Muñoz, and Alberto Suárez, *Member*, *IEEE*

Abstract—The global prediction of a homogeneous ensemble of classifiers generated in independent applications of a randomized learning algorithm on a fixed training set is analyzed within a Bayesian framework. Assuming that majority voting is used, it is possible to estimate with a given confidence level the prediction of the complete ensemble by querying only a subset of classifiers. For a particular instance that needs to be classified, the polling of ensemble classifiers can be halted when the probability that the predicted class will not change when taking into account the remaining votes is above the specified confidence level. Experiments on a collection of benchmark classification problems using representative parallel ensembles, such as bagging and random forests, confirm the validity of the analysis and demonstrate the effectiveness of the instance-based ensemble pruning method proposed.

Index Terms—Ensemble learning, bagging, random forests, ensemble pruning, instance-based pruning, Polya urn.

1 Introduction

THERE is a large body of theoretical and empirical research showing that combining the predictions of a collection of classifiers can be an effective strategy to improve generalization performance [1], [2], [3], [4], [5], [6], [7], [8]. Homogeneous ensembles composed of independent classifiers generally use equally weighted voting to produce a final class prediction. Given an unlabeled instance, the usual procedure is to query all classifiers in the ensemble and then output the majority class. In this work, we show that it is possible to estimate the outcome of the voting process with a specified confidence level by polling only a subset of the classifiers in the ensemble. The procedure is based on estimating the probability distribution of the remaining class predictions using the decisions made by the classifiers that have already been queried. This probability distribution is a Polya urn model with $c = 1$, for which explicit formulas exist. Using this distribution, it is possible to estimate the probability that the predicted class will not change when taking into account the remaining votes. Assuming that there is some tolerance to disagreements between the predictions of the subensemble and of the complete ensemble, the querying process can be stopped when this probability exceeds the specified confidence level. The final classification is the aggregate decision of the polled classifiers only.

This type of instance-based (IB) pruning can be used to improve the classification efficiency of any homogeneous ensemble provided that its members are generated on independent applications of a randomized learning algorithm on the same training data, and that majority voting is used. Examples of ensembles of this kind can be found in [2], [4], [5], [6], [7]. In this work, we present results for bagging [2] and random forests (RF) [4], which are two representative ensemble learning algorithms. Experiments on a variety of classification problems show that the rate of disagreement in the class label predictions by the complete ensemble and by the polled subensembles are in the vicinity of the specified confidence level. Furthermore, only small differences in generalization performance are found. Using this procedure, only a subset of the predictors in the ensemble needs to be queried. As a consequence, the classification process is significantly sped up.

The statistical description of the evolution of the prediction error in majority voting algorithms as the number of classifiers in the ensemble increases is already presented in [9]. Similar analyses are reproduced in later works [6], [10], [11]. However, to the knowledge of the authors, the statistical properties of the aggregation process have not been used as a basis for ensemble pruning. Fan et al. [12], [13] employ a procedure to speed up classification in ensembles based on ideas similar to the ones used in this work. In these articles, the goal is to minimize the number of classifiers needed for prediction in cost-sensitive applications. For this purpose, it is assumed that the distribution of the discrepancies between the predictions of the subensemble and of the complete ensemble is normal, and that the parameters of this distribution can be estimated on the training data. The method proposed by Fan et al. assumes that the individual classifiers output continuous probabilities and therefore cannot be applied without modifications to majority voting. A more serious difficulty is that, in the classification problems investigated, the discrepancies between the partial and the complete ensemble class label predictions tend to be smaller for the training set than for independent test sets. This means that, in general, the estimates on the training data are biased. On average, for the same confidence level, fewer classifiers need to be queried for training instances than for instances outside of the training set.

2 Statistical Instance-Based Ensemble Pruning

Consider an ensemble of T classifiers. Assuming that majority voting is used to combine the decisions of the individual learners, the class predicted by the ensemble for an unlabeled instance, characterized by the vector of attributes $\mathbf{x}$, is

$$\underset{y}{arg\,max} \sum_{i=1}^{T} \mathbb{I}(h_i(\mathbf{x}) = y), y \in \mathcal{Y}, \quad (1)$$

where $h_i(\mathbf{x})$ is the class prediction of the ith member of the ensemble, $\mathbb{I}$ is an indicator function, and $\mathcal{Y} = \{y_1, \ldots, y_l\}$ is the set of possible class labels.

This voting scheme is usually implemented by first computing the outputs of all the classifiers in the ensemble and then finding the majority class. However, for most instances, polling all classifiers is not necessary to determine the final prediction: If, in the first t queries $(t < T)$, the difference of votes between the majority class and the second most voted class is larger than $T - t$, then the class prediction cannot be changed by the remaining (still unknown) votes. In particular, for binary classification problems, the querying process can be stopped when more than 50 percent of the ensemble members agree in their prediction. This behavior is more likely to occur for instances that are classified by the ensemble with a large margin [14].

The querying process can actually be stopped earlier if one is willing to accept that a specified (small) fraction of the class labels predicted by the polled subensemble are in disagreement with the prediction of the complete ensemble. With regard to the generalization performance, the disagreement rate is an upper bound for the differences between the error rate of the polled subensemble and of the complete ensemble. In practice, if the changes in class labels affect both correctly labeled examples and misclassified

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examples in approximately equal numbers, the differences in error rates should be much smaller than this upper bound.

Consider an ensemble in which the individual classifiers are generated in independent executions of a randomized learning algorithm using a fixed set of labeled examples. That is, the ensemble members are independent when conditioned to these training data. Under these conditions, the classification process of an arbitrary instance $\mathbf{x}$ can be described in terms of the probability vector

$$\mathbf{p}(\mathbf{x}) = \{p_1(\mathbf{x}), p_2(\mathbf{x}), \ldots, p_l(\mathbf{x})\}, \quad \sum_{i=1}^{l} p_i(\mathbf{x}) = 1, \tag{2}$$

where $p_i(\mathbf{x}) \geq 0$ is the probability that instance $\mathbf{x}$ is assigned class label y_i by an individual classifier in the ensemble. The values of these probabilities are unknown and depend on the ensemble learning algorithm, on the particular classification problem and on the current instance $\mathbf{x}$. The classification of this instance by an ensemble of arbitrary size t can be seen as a sequence of t independent trials, where each trial has a fixed number of possible outcomes in the set $\mathcal{Y}$. Given $\mathbf{p}(\mathbf{x})$, the probability distribution of the possible outcomes of these t experiments is a multinomial distribution

$$\mathcal{P}(\mathbf{t}|\mathbf{p}) = \frac{t!}{t_1!t_2!\ldots t_l!} p_1^{t_1} p_2^{t_2} \ldots p_l^{t_l}, \tag{3}$$

where $\mathbf{t} = \{t_1, t_2, \ldots, t_l;\ \sum_{i=1}^{l} t_i = t\}$ and t_i is the number of classifiers that predict class y_i for the current instance $\mathbf{x}$. The class predicted by the ensemble of size t is

$$\hat{k}_{\mathbf{t}} = \underset{i}{arg\,max}\{t_i; i = 1, 2, \ldots, l\}. \tag{4}$$

To simplify the notation, the dependence on $\mathbf{x}$ of $\mathbf{p}$ has been dropped from these and the following expressions.

Under the assumption that there is no information about the a priori distribution of classes and that all possible values of the probability vector in (3) are equally likely, $\mathcal{P}(\mathbf{p})$ is a uniform distribution. This prior can be plugged into Bayes theorem to compute the posterior probability distribution of the probability vector $\mathbf{p}$, given the observed values of $\mathbf{t}$,

$$\begin{aligned} \mathcal{P}(\mathbf{p}|\mathbf{t}) &= \frac{\mathcal{P}(\mathbf{t}|\mathbf{p})\mathcal{P}(\mathbf{p})}{\mathcal{P}(\mathbf{t})} \\ &= \frac{p_1^{t_1} p_2^{t_2} \ldots p_l^{t_l}}{\int_{\mathcal{D}} p_1^{t_1} p_2^{t_2} \ldots p_l^{t_l} dp_1 dp_2 \ldots dp_l} \\ &= \frac{\Gamma(\sum_{i=1}^{l} t_i + l)}{\prod_{i=1}^{l} \Gamma(t_i + 1)} p_1^{t_1} p_2^{t_2} \ldots p_l^{t_l}, \end{aligned} \tag{5}$$

where the region of integration, $\mathcal{D}$, is the space of l-dimensional probability vectors, and $\Gamma(z)$ is the gamma function. The posterior for $\mathbf{p}$ is a Dirichlet distribution of order l with parameters $(t_1 + 1, t_2 + 1, \ldots, t_l + 1)$ [15].

Our goal is to determine how the prediction of an ensemble of size T can be estimated with a certain level of confidence after only $t < T$ of its classifiers have been queried. Consider the vector of integers that summarizes the predictions of the complete ensemble of size T

$$\mathbf{T} \equiv \left\{ T_1, T_2, \ldots, T_l; \sum_{i=1}^{l} T_i = T \right\}, \tag{6}$$

where T_i is the number of members in the ensemble that predict class i for the instance to be classified. Assuming that $\mathbf{t} \equiv \{t_1, t_2, \ldots, t_l;\ \sum_{i=1}^{l} t_i = t;\ t_i \leq T_i, i = 1, 2, \ldots, l\}$, the vector that stores the predictions of the first $t \leq T$ classifiers in the ensemble, is known, the probability distribution for $\mathbf{T}$ is

$$\begin{aligned} \mathcal{P}(\mathbf{T}|\mathbf{t}) &= \int_{\mathcal{D}} \mathcal{P}(\mathbf{T} - \mathbf{t}|\mathbf{p})\mathcal{P}(\mathbf{p}|\mathbf{t}) d\mathbf{p} \\ &= \frac{(T-t)!}{\prod_{i=1}^{l}(T_i - t_i)!} \frac{\prod_{i=1}^{l}(t_i+1)_{T_i - t_i}}{(t+l)_{T-t}}, \end{aligned} \tag{7}$$

where $\{T_i \geq t_i,\ i = 1, 2, \ldots, l\}$, and $(a)_n = a(a+1)\cdots(a+n-1)$ is the Pocchammer symbol, or rising factorial, with a and n nonnegative integers [16]. Note that $\mathbf{T} - \mathbf{t}$ is independent of $\mathbf{t}$ given $\mathbf{p}$ because classifiers are built on independent executions of the learning algorithm.

Expression (7) describes a Polya urn model with $t_i + 1$ balls of color i, $i = 1, 2, \ldots, l$ and $c = 1$. Hence, there are a total of $t + l$ balls of l different colors. An experiment with repeated extractions from the urn is made. In each extraction, a ball from the urn is selected at random and then returned to the urn together with c other balls of the same color. For $c = 1$, the probability of extracting $\mathbf{T} - \mathbf{t}$ balls of each color after $T - t$ extractions is precisely (7) [17].

The probability that the class labels predicted by the subensemble of size $t < T$ and by the complete ensemble of size T coincide is

$$\tilde{\mathcal{P}}(\mathbf{t}, T) = \sum_{\mathbf{T}}^{*} \frac{(T-t)!}{\prod_{i=1}^{l}(T_i - t_i)!} \frac{\prod_{i=1}^{l}(t_i+1)_{T_i - t_i}}{(t+l)_{T-t}}, \tag{8}$$

where the asterisk indicates that the summation runs over all values of $\mathbf{T}$ such that $\sum_{i=1}^{l} T_i = T$, $\{T_i \geq t_i,\ i = 1, 2, \ldots, l\}$, and $T_{\hat{k}_{\mathbf{t}}} > T_j$ for $j \neq \hat{k}_{\mathbf{t}}$, where $\hat{k}_{\mathbf{t}}$ is the majority class after querying the first t classifiers. If we are willing to accept that the coincidence between these two predictions is not necessarily certain, but occurs with a high confidence level π, (8) can be used to determine the number of classifiers needed to estimate the prediction of the complete ensemble: the querying process can be halted after t^* classifiers have been queried, where the vector of class predictions of the current subensemble $\mathbf{t}^*$ is such that $\tilde{\mathcal{P}}(\mathbf{t}^*, T) \geq \pi$.

For $T > l$, the cost of computing (8) is exponential in the number of different class labels l. In consequence, for typical ensemble sizes $(T \approx 100)$, the evaluation becomes costly for problems with a large number of classes. However, in binary classification problems, expression (8) can be readily computed. For fixed values T and π, the table for $\tilde{\mathcal{P}}(\mathbf{t}, T)$, which has $(T+1)(T+2)/2$ elements, can be replaced by an equivalent table of size T. The tth entry of this table is $t^*(t; T, \pi)$, the minimum number of observations of the majority class label required to obtain the same prediction as the complete ensemble with a confidence level π, when t classifiers have been queried. Thus, when classifying an instance, polling can be stopped when t_{prune} classifiers have been queried and the number of classifiers that predict the majority class is $t^*(t_{prune}; T, \pi)$. Fig. 1 displays the critical values $t^*(t; T, \pi)/t$ for an ensemble of $T = 101$ classifiers and a confidence level $\pi = 99\%$. An ensemble with an odd number of classifiers is used to avoid ties in the majority voting algorithm. In particular, it is necessary that the first six classifiers queried agree to have a probability above 99 percent that this classification is the same as the one given by the complete ensemble. The first few values of $t^*(t)/t$ are 6/6, 7/7, 8/8, 8/9, 9/10, 10/11, 10/12, 11/13, ... The nonmonotonic behavior observed in the graph arises from the fact that both $t^*(t)$ and t are integers. As the number of available observations t increases, the value of $t^*(t)$ approaches from above the limit for the complete ensemble $t^*(t = 101; T = 101, \pi) = 51$.

Despite the fact that the computation of (8) can be costly for problems with a large number of class labels, it is a one-time calculation. The calculation of $\tilde{\mathcal{P}}(\mathbf{t}, T)$ relies solely on the statistical properties of the majority voting process. In particular, these probabilities do not depend on the details of the classification task to be solved. For a given value of π, they need to be calculated only

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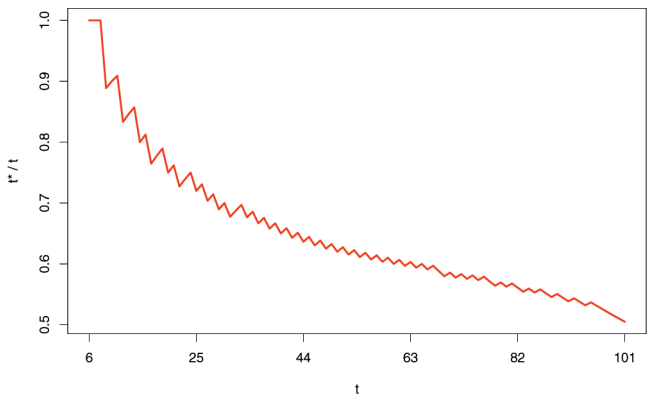


Fig. 1. Minimum value of the fraction of classifiers that need to agree in their class predictions to guarantee that, for a binary classification problem, the majority class after polling t classifiers is the same as the one predicted by the complete ensemble of size $T = 101$, with probability $\pi = 99\%$, as a function of t, the number of classifiers whose prediction is known.

once and then stored in a lookup table of size $(T+1)_l/l!$. Henceforth, they can be used for *any* classification task. This means that the overhead of the IB ensemble pruning procedure is negligible because it requires only retrieving values stored in a precomputed lookup table. As a consequence, the pruning rates obtained directly translate into a speedup of the classification process.

3 Experiments

A series of experiments are performed to illustrate the theoretical analysis of the ensemble pruning procedure described in the previous section. These experiments are also used to assess the efficiency of IB pruning in reducing the expected prediction time in a collection of benchmark classification problems.

A first batch of experiments is carried out in a simple binary classification task that can be analyzed in detail. The problem consists in predicting which points within the square $[-1,1]^2$ are also inside a circumference of diameter $\sqrt{2}$, centered at the origin. The Cartesian coordinates of the points $\mathbf{x} = (x_1, x_2)$ are used as attributes for prediction. Instances are sampled uniformly in the square $[-1,1]^2$. The class assigned to each instance is 1 if $x_1^2 + x_2^2 \geq 0.5$, 0 otherwise. To make the problem more realistic, the class labels in 5 percent of the training instances, selected at random, are changed. An RF is built by learning $T = 101$ trees from a fixed set of 100 labeled instances [4]. The ensemble is tested on a set of 10,000 points located on a regular grid within the square $[-1,1]^2$. For each instance, querying is stopped when the prediction of the subensemble of polled classifiers is expected to coincide with the prediction of the complete ensemble at a confidence level $\pi = 99\%$. Fig. 2 displays the results of these experiments, averaged over 200 independent realizations of the training set. In the graph in Fig. 2a, different shades of gray are used to denote the average fraction of trees that are needed to output a decision with the specified level of confidence for instances in that particular region of attribute space. Lighter shades of gray correspond to smaller subensembles. Regions where it is necessary to query all the classifiers in the original ensemble are drawn in black. Contour curves are also displayed for reference. The optimal decision boundary of the classification problem is drawn as a white circumference centered at the origin. Data instances that are far from the optimal decision boundary generally require querying fewer classifiers. In contrast, the amount of disagreement among classifiers is larger for data instances close to the decision boundary. In consequence, more queries are needed to produce a decision at the same confidence level. Nevertheless, even in this region, the classification of most of the instances requires knowing the prediction of only a subset of the learners in the ensemble. The histogram in Fig. 2b displays the average fraction of test examples whose class label can be predicted by querying a given number of classifiers in the ensemble. Note the bimodal form of the histogram, where the mode around 30 corresponds to instances close to the classification boundary. For this problem, on average, only 15 percent of the original classifiers need be polled to output a decision that coincides with the prediction of the complete ensemble with a probability above 99 percent.

The performance of IB pruning is further evaluated in a set of benchmark binary classification problems, including data sets from the UCI Repository [18], the synthetic problems *Twonorm*, *Ringnorm*, and *Threenorm* from [19], *task 1* from the 2008 UC San Diego

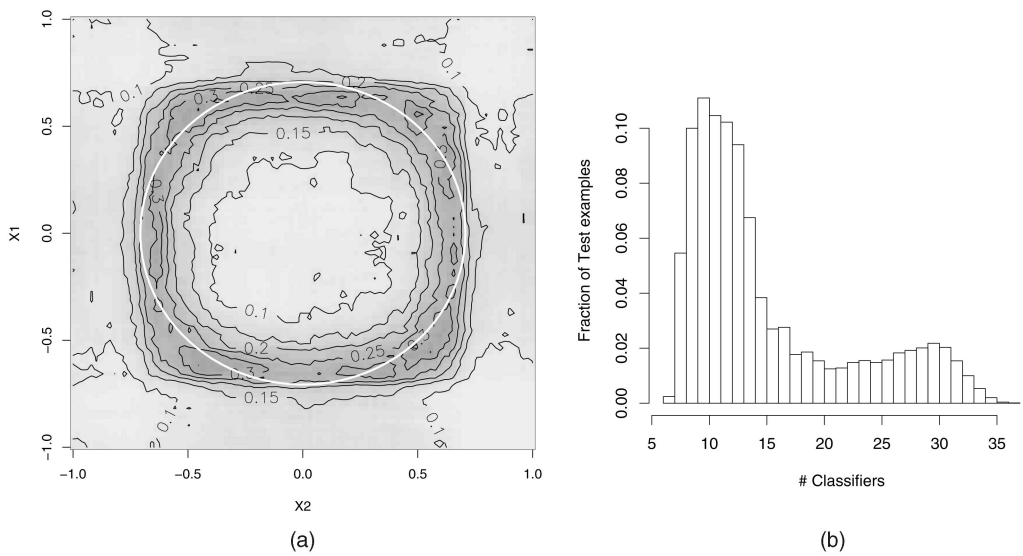


Fig. 2. Average numbers of trees used by bagging with IB pruning ($\pi = 99\%$) to estimate the class labels of test instances in the square $[-1,1]^2$. (a) In the graph, different shades of gray are used to indicate the fraction of trees that are needed on average to predict an instance located in that region of attribute space. (b) The histogram displays the fraction of test instances whose class can be predicted querying a given number of classifiers in the ensemble.

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TABLE 1
Disagreement Rates in Percent between Complete and Pruned Ensembles ($\pi = 99\%$)

	Bagging		Random Forests	
Problem	IB-Bag	FR-Bag	IB-RF	FR-RF
australian	0.0 ± 0.0	0.4 ± 1.4	0.4 ± 0.4	4.3 ± 1.5
boston	0.2 ± 0.4	3.0 ± 2.4	0.4 ± 0.5	4.9 ± 1.6
breast	0.1 ± 0.2	1.2 ± 0.8	0.1 ± 0.2	1.3 ± 0.7
gisette	0.1 ± 0.1	1.3 ± 0.4	0.1 ± 0.1	2.0 ± 0.4
heart	0.3 ± 1.0	4.2 ± 3.1	0.6 ± 0.8	5.3 ± 2.3
hepatitis	0.0 ± 0.2	1.2 ± 2.9	0.4 ± 0.8	5.2 ± 3.1
ionosphere	0.1 ± 0.5	2.2 ± 2.0	0.1 ± 0.3	2.5 ± 1.4
liver	0.7 ± 1.4	6.0 ± 4.0	1.0 ± 1.0	7.9 ± 2.8
magic	0.1 ± 0.1	2.2 ± 0.3	0.3 ± 0.1	3.9 ± 0.3
microarray	0.2 ± 1.8	2.1 ± 8.0	0.8 ± 3.0	6.6 ± 9.1
mushroom	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0
pima	0.3 ± 0.6	3.3 ± 2.1	0.6 ± 0.5	6.0 ± 1.7
ringnorm	0.5 ± 0.3	5.7 ± 1.2	0.4 ± 0.2	4.9 ± 0.9
sonar	0.2 ± 0.6	4.3 ± 3.4	0.9 ± 1.3	7.9 ± 3.2
spam	0.1 ± 0.1	1.7 ± 0.4	0.2 ± 0.1	2.3 ± 0.4
threenorm	1.0 ± 0.5	7.6 ± 1.5	1.0 ± 0.3	7.9 ± 1.0
tic-tac-toe	0.1 ± 0.3	2.2 ± 0.9	0.3 ± 0.3	3.3 ± 1.2
twonorm	0.6 ± 0.3	5.4 ± 1.0	0.4 ± 0.2	4.3 ± 0.7
task 1	0.0 ± 0.0	1.1 ± 0.2	0.1 ± 0.0	1.9 ± 0.1
votes	0.0 ± 0.1	0.3 ± 0.7	0.1 ± 0.2	1.5 ± 1.2

Data Mining Contest (see http://mill.ucsd.edu/), which is a binary classification problem with a very unbalanced class distribution, and the *microarray* data set described in [20], where the goal is to determine whether a particular patient carries the BRCA1 mutation or not. The Boston problem from UCI is transformed into a binary classification task by discriminating between houses worth more than $21,000 and the rest. In the problem *Gisette*, the 20 features that are correlated the most with the class label are used for prediction. The results reported are averages over 100 realizations of each problem. These correspond to either different simulations of the data in the problems where a method to generate labeled instances is available (300 instances for training, 1,000 for testing) or to random partitions into train (2/3 of the available data) and test sets (the remaining 1/3) in the remaining sets. In each experiment, a bagging ensemble and an RF of 101 elements are built. The predictions of the complete ensembles are compared with the predictions of the corresponding ensembles pruned using IB pruning with $\pi = 99\%$. Note that it is not necessary to query all classifiers to determine the class prediction of the complete ensemble for a given instance with certainty. It suffices to query classifiers until the count for the majority class is larger than 50 percent of the number of classifiers in the original ensemble. Therefore, in an ensemble of size $T = 101$, polling can be stopped without loss of relevant information when 51 classifiers agree in their class prediction. This corresponds to IB pruning with $\pi = 100\%$ and provides a lower bound for the pruning rates that can be achieved.

One can ask whether it is really necessary to use a different stopping point for each instance or whether a fixed pruning rate is sufficient to achieve predictions that are close to those of the complete ensemble. To investigate this issue, the results of an ensemble that queries a subset of fixed size, independently of the instance that is being classified, are reported as well. The number of classifiers queried in fixed-rate (FR) pruning is the ceiling of the average number of classifiers obtained by IB pruning with $\pi = 99\%$.

Table 1 shows the percentage of disagreement in the test set between the class prediction of the original complete ensembles and the pruned ones for bagging (columns 2 and 3) and for RF (columns 4 and 5). The results are given in the format *average* ± *standard deviation*. For IB pruning, the average disagreement rates are in the vicinity of the expected value $100 - \pi = 1\%$, and generally below it. In contrast, when a fixed ensemble size is used (i.e., in FR-Bag and FR-RF), these rates need not be close to this target and can be rather large. The disagreement rates are in fact an upper bound for the differences in classification error. Even when large disagreement rates are obtain, similar error rates can be

TABLE 2
Error Rates and Number of Trees in Pruned Ensembles for Bagging

		Test error		Number of trees		speed-up
Problem	Bagging	IB-Bag	FR-Bag	$\pi = 100\%$	$\pi = 99\%$	factor
australian	14.5±1.9	14.5±1.9	14.5±2.0	52.5±1.1	7.0±1.5	7.7±1.0
boston	17.1±3.0	17.1±3.0	17.1±2.9	57.4±1.5	12.4±2.3	4.8±0.8
breast	5.0±1.5	5.0±1.5	5.1±1.6	53.9±0.5	8.6±0.8	6.3±0.5
gisette	9.1±0.6	9.1±0.6	9.2±0.7	53.5±0.2	8.4±0.4	6.4±0.3
heart	20.8±4.7	20.8±4.6	21.2±4.5	63.6±2.3	19.1±3.9	3.5±0.7
hepatitis	21.2±4.4	21.3±4.4	21.1±4.6	55.2±2.0	9.1±3.0	6.5±1.5
ionosphere	9.0±2.8	9.0±2.9	9.5±3.0	56.1±1.4	10.5±2.1	5.5±1.0
liver	33.9±5.6	34.0±5.7	34.6±5.9	68.4±3.1	23.2±5.6	3.1±0.8
magic	14.3±0.5	14.3±0.5	14.5±0.5	55.6±0.2	10.5±0.3	5.3±0.1
microarray	30.2±11.3	30.2±11.4	30.0±12.0	60.0±6.4	13.2±9.4	6.0±2.4
mushroom	0.0±0.1	0.0±0.1	0.0±0.1	51.0±0.0	6.0±0.0	8.5±0.0
pima	25.4±2.6	25.4±2.7	25.7±2.6	58.9±1.6	13.5±2.4	4.5±0.9
ringnorm	12.2±3.0	12.3±3.0	13.3±2.8	68.3±1.3	23.7±1.8	2.9±0.2
sonar	25.5±4.4	25.5±4.4	25.8±4.2	65.2±3.9	19.2±5.5	3.6±1.0
spam	7.6±0.7	7.6±0.7	7.9±0.7	54.6±0.2	9.4±0.4	5.8±0.2
threenorm	23.9±3.4	24.1±3.3	24.9±3.2	75.1±1.6	32.9±2.9	2.3±0.2
tic-tac-toe	2.6±1.1	2.7±1.1	3.9±1.3	57.8±0.6	11.6±1.0	5.0±0.4
twonorm	9.8±2.9	10.0±2.9	11.3±2.7	69.0±0.6	24.2±1.2	2.9±0.1
task 1	6.1±0.2	6.1±0.2	6.2±0.2	52.8±0.1	7.7±0.1	6.8±0.1
votes	4.6±1.3	4.6±1.3	4.7±1.4	51.7±0.3	6.5±0.5	8.0±0.5

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TABLE 3
Error Rates and Number of Trees in Pruned Ensembles for Random Forests

	Test error			Number of trees		speed-up
Problem	RF	IB-RF	FR-RF	$\pi = 100\%$	$\pi = 99\%$	factor
australian	13.2±1.9	13.3±1.9	13.7±1.7	63.1±0.9	17.4±1.6	3.7± 0.3
boston	13.4±2.2	13.4±2.2	14.0±2.5	62.6±1.0	17.9±1.7	3.5± 0.3
breast	3.2±0.9	3.2±0.9	3.9±1.1	54.3±0.5	8.9±0.7	6.1± 0.4
gisette	7.8±0.5	7.8±0.5	8.2±0.5	55.8±0.2	10.3±0.3	5.4± 0.2
heart	17.7±3.5	17.7±3.6	18.5±3.7	67.5±1.4	22.5±2.6	3.0± 0.3
hepatitis	15.9±4.3	16.0±4.3	16.2±4.2	64.0±2.2	19.6±3.6	3.4± 0.6
ionosphere	6.8±2.1	6.8±2.1	7.5±2.3	58.8±1.0	12.6±1.7	4.7± 0.6
liver	28.7±3.7	28.8±3.6	30.2±3.7	76.0±1.5	34.6±3.0	2.2± 0.2
magic	12.2±0.4	12.2±0.4	12.9±0.4	61.7±0.2	15.9±0.3	3.9± 0.1
microarray	26.8±13.6	26.5±14.1	26.9±14.4	74.6±4.8	31.1±10.7	2.7± 0.9
mushroom	0.0±0.0	0.0±0.0	0.0±0.0	51.1±0.0	6.1±0.0	8.4± 0.0
pima	24.3±2.0	24.3±2.1	25.0±2.2	69.5±0.9	25.4±1.9	2.7± 0.2
ringnorm	6.7±1.1	6.8±1.1	8.5±1.2	68.7±0.9	22.9±1.2	3.0± 0.1
sonar	18.8±4.5	19.0±4.5	19.6±4.8	74.4±1.4	32.7±3.3	2.3± 0.2
spam	5.1±0.5	5.1±0.5	5.7±0.6	57.4±0.3	11.5±0.4	5.0± 0.2
threenorm	17.2±1.5	17.3±1.5	18.9±1.6	76.9±0.7	35.4±1.3	2.2± 0.1
tic-tac-toe	2.4±0.9	2.6±0.9	4.5±1.4	64.9±0.6	18.0±1.3	3.6± 0.2
twonorm	4.4±0.8	4.5±0.8	6.4±0.8	67.2±0.4	20.9±0.7	3.2± 0.1
task 1	5.3±0.2	5.3±0.2	5.6±0.2	55.6±0.1	10.1±0.1	5.5± 0.1
votes	3.8±1.4	3.9±1.3	4.4±1.5	55.1±0.8	9.3±1.1	6.0± 0.6

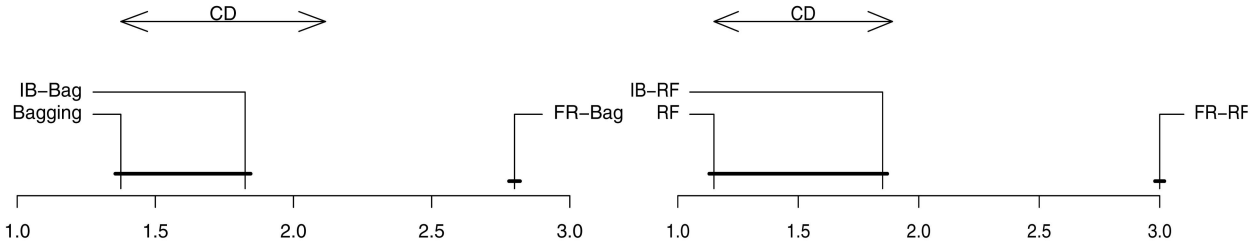


Fig. 3. (Left) Average ranks in classification performance for complete bagging, bagging ensembles pruned using IB pruning with $\pi = 99\%$ (IB-Bag), and bagging ensembles pruned with a fixed rate (FR-Bag). (Right) Average ranks in classification performance for RF, RF pruned using IB pruning with $\pi = 99\%$ (IB-RF), and RF pruned with a fixed rate (FR-RF). The significance of the differences in average rank of the different ensembles is determined using a Nemenyi test. The critical value for the differences (CD) is given by the length of the segment plotted at the top.

found if the changes in class label affect approximately the same numbers of correctly and incorrectly classified instances.

Tables 2 and 3 display the results of experiments for the error rate and size of the pruned ensembles in bagging and RF, respectively. Columns 2 to 4 of these tables display the test error of the complete ensembles, of ensembles pruned with IB pruning using $\pi = 99\%$ (*IB-Bag* in Table 2 and *IB-RF* in Table 3), and of ensembles pruned with a fixed rate (*FR-Bag* in Table 2 and *FR-RF* in Table 3), respectively. These results confirm that RF generally outperforms bagging [7]. The differences in performance with respect to the original ensemble when IB pruning is used are fairly small. The error rates for FR pruning are generally larger, especially in RF.

The average number of trees that needs to be queried to determine the complete ensemble prediction at a confidence level π is tabulated in the fifth and sixth columns of Tables 2 and 3. Column 5 corresponds to IB-pruned ensembles with $\pi = 100\%$, which means that these predictions coincide with those of the complete ensemble. Column 6 corresponds to IB pruning with $\pi = 99\%$ (IB-Bag and IB-FR). Since the overhead to determine when to stop querying is negligible (one simply needs to compare the majority class count with a critical value, stored in a precalculated lookup table), the reduction in ensemble size by pruning directly entails a reduction in the time needed to classify a given instance. Column 7 displays the factor by which the classification time is reduced when statistical IB pruning is used with $\pi = 99\%$ instead of $\pi = 100\%$. This speed-up factor in classification varies between $\approx$2 and $\approx$8.5, depending on the problem considered. Problems that are simple for classification trees, such as the *Mushroom* problem, need, on average, very few classifiers to achieve a good prediction. In contrast, for *Threenorm*, which is a difficult task for classification trees, a large number of trees are required to reach a decision. In general, RF need to query more trees than bagging. This is a consequence of the larger variability introduced by the RF algorithm when building the individual classifiers. As a result, the learners in RF tend to be more diverse than those generated in bagging and typically agree less frequently in their predictions.

The performance on the different ensembles is compared using the statistical methodology proposed in [21]. A Friedman test provides evidence of statistically significant differences in performances at a 95 percent confidence level both for bagging and for RF. A Nemenyi test at a 5 percent significance level is carried out to compare the performance of the different ensembles in terms of their average ranks in the classification problems investigated. The results of this test are displayed in Fig. 3: for bagging (Fig. 3 left) and for RF (Fig. 3 right). They show that ensembles pruned using

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an FR perform significantly worse than those pruned by means of IB pruning. On the other hand, the empirical evidence is not sufficient to determine whether the differences in performance between the complete ensembles and the corresponding sub-ensembles selected with IB pruning are statistically significant. These results underscore the importance of determining the appropriate pruning rate for each specific instance.

4 Conclusions

In this paper, we have proposed a statistical method for estimating how many classifiers in an ensemble need to be queried to determine the prediction of the complete ensemble with a specified confidence level. For a given unlabeled instance, the classification process is assumed to proceed by querying a single classifier at a time. Starting from a uniform prior, Bayes theorem is used to update the probabilities of the complete ensemble predicting a given class label using the evidence available when the output of an additional classifier in the ensemble becomes known. The polling process is stopped when the probability that the ensemble prediction will change after having queried the remaining classifiers is below a specified level. The number of queries required for this estimation depends on the instance that is being classified. For some instances, where the ensemble members tend to agree in their predictions, only a few classifiers need to be polled to gather sufficient evidence about the final prediction. Other instances, typically those that are close to the classification boundary, require a larger number of classifiers. Given that the time needed to determine when to stop querying is small, IB pruning directly translates into a speedup of the classification process.

Because it relies solely on the statistical properties of the aggregation process, the type of IB pruning proposed can be applied to any classification task, using any randomized ensemble learning algorithm, provided that the individual classifiers in the ensemble are of the same type, that they are built independently of each other, and that their decisions are combined via majority voting.

An empirical investigation using two parallel ensemble creation methods, bagging and RF, demonstrates that this statistical IB method for ensemble pruning is sound and effective. In particular, the average number of classifiers needed for classification is substantially reduced in all the problems investigated. Furthermore, the generalization performance of IB-pruned ensembles with $\pi = 99\%$ is comparable to the corresponding complete ensembles.

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References

[1] Y. Freund and R.E. Schapire, "A Decision-Theoretic Generalization of On-Line Learning and an Application to Boosting," *Proc. Second European Conf. Computational Learning Theory*, pp. 23-37, 1995.

[2] L. Breiman, "Bagging Predictors," *Machine Learning*, vol. 24, no. 2, pp. 123-140, 1996.

[3] T.G. Dietterich, "An Experimental Comparison of Three Methods for Constructing Ensembles of Decision Trees: Bagging, Boosting, and Randomization," *Machine Learning*, vol. 40, no. 2, pp. 139-157, 2000.

[4] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5-32, 2001.

[5] P. Buhlmann, "Bagging, Subagging and Bragging for Improving Some Prediction Algorithms," *Recent Advances and Trends in Nonparametric Statistics*, M. Akritas and D. Politis, eds., pp. 19-34, 2003.

[6] G. Martínez-Muñoz and A. Suárez, "Switching Class Labels to Generate Classification Ensembles," *Pattern Recognition*, vol. 38, no. 10, pp. 1483-1494, 2005.

[7] J.J. Rodriguez, L.I. Kuncheva, and C.J. Alonso, "Rotation Forest: A New Classifier Ensemble Method," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 28, no. 10, pp. 1619-1630, Oct. 2006.

[8] R.E. Banfield, L.O. Hall, K.W. Bowyer, and W.P. Kegelmeyer, "A Comparison of Decision Tree Ensemble Creation Techniques," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 29, no. 1, pp. 173-180, Jan. 2007.

[9] L.K. Hansen and P. Salamon, "Neural Network Ensembles," *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 12, pp. 993-1001, 1990.

[10] L. Lam and C. Suen, "Application of Majority Voting to Pattern Recognition: An Analysis of Its Behavior and Performance," *IEEE Trans. Systems, Man, and Cybernetics*, vol. 27, no. 5, pp. 553-568, 1997.

[11] R. Esposito and L. Saitta, "Monte Carlo Theory as an Explanation of Bagging and Boosting," *Proc. 18th Int'l Joint Conf. Artificial Intelligence*, pp. 499-504, 2003.

[12] W. Fan, F. Chu, H. Wang, and P.S. Yu, "Pruning and Dynamic Scheduling of Cost-Sensitive Ensembles," *Proc. 18th Nat'l Conf. Artificial Intelligence*, pp. 146-151, 2002.

[13] H. Wang, W. Fan, P.S. Yu, and J. Han, "Mining Concept-Drifting Data Streams Using Ensemble Classifiers," *Proc. Ninth ACM SIGKDD Int'l Conf. Knowledge Discovery and Data Mining*, pp. 226-235, 2003.

[14] R. Schapire, Y. Freund, P. Bartlett, and W. Lee, "Boosting the Margin: A New Explanation for the Effectiveness of Voting Methods," *The Annals of Statistics*, vol. 12, no. 5, pp. 1651-1686, 1998.

[15] C.M. Bishop, *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Springer, Aug. 2006.

[16] M. Abramowitz and I.A. Stegun, *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. Dover, 1964.

[17] N.L. Johnson, S. Kotz, and N. Balakrishnan, *Discrete Multivariate Distributions*. John Wiley & Sons, 1997.

[18] A. Asuncion and D. Newman, "UCI Machine Learning Repository," http://www.ics.uci.edu/mlearn/MLRepository.html, 2007.

[19] L. Breiman, "Bias, Variance, and Arcing Classifiers," Technical Report 460, Statistics Dept., Univ. of California, Berkeley, 1996.

[20] I. Hedenfalk, D. Duggan, Y. Chen, M. Radmacher, M. Bittner, R. Simon, P. Meltzer, B. Gusterson, M. Esteller, O.P. Kallioniemi, B. Wilfond, A. Borg, and J. Trent, "Gene-Expression Profiles in Hereditary Breast Cancer," *The New England J. Medicine*, vol. 344, no. 8, pp. 539-548, 2001.

[21] J. Demšar, "Statistical Comparisons of Classifiers over Multiple Data Sets," *J. Machine Learning Research*, vol. 7, pp. 1-30, 2006.

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